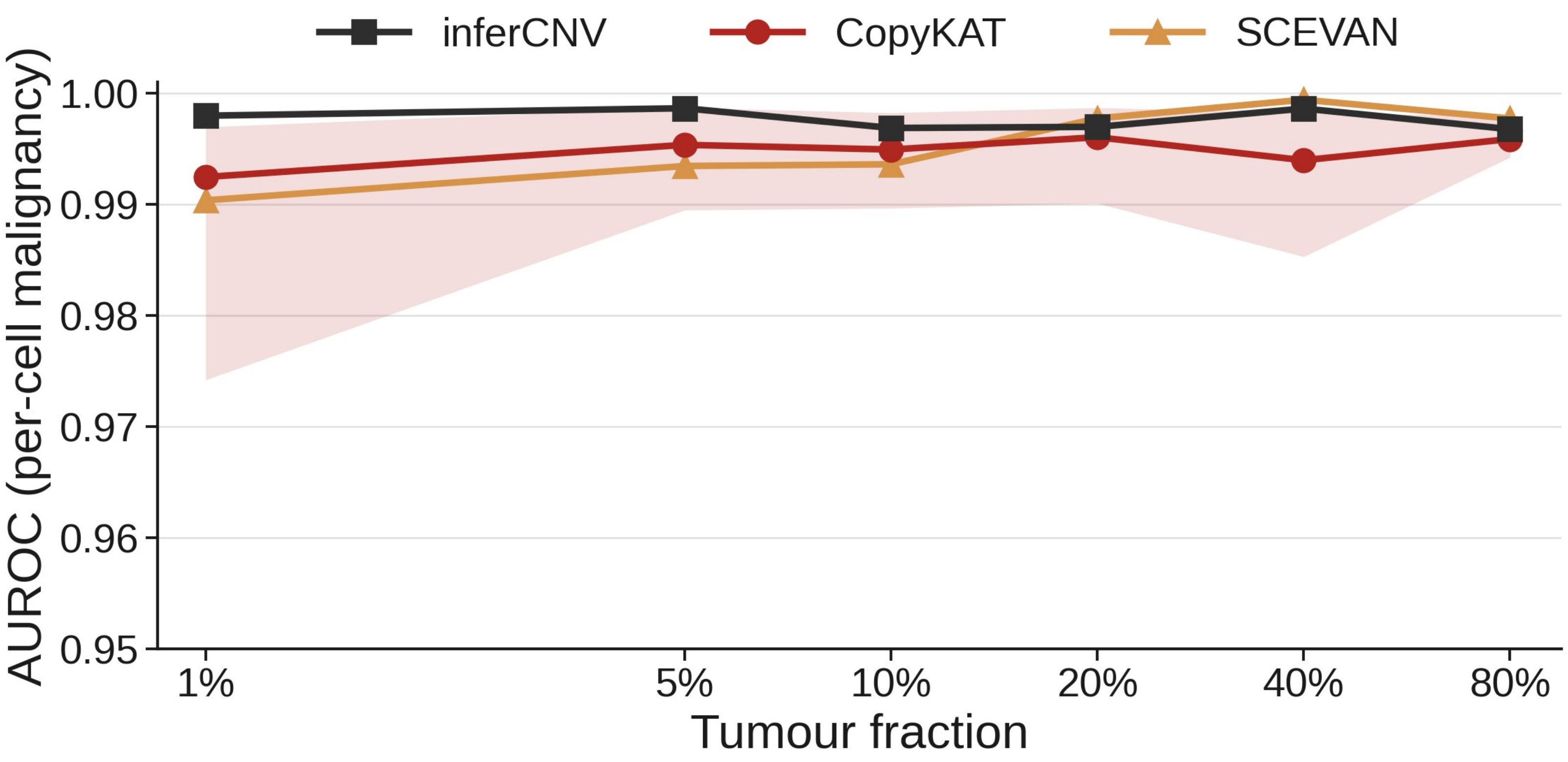
Discrimination holds across tumour purity

Purity-resolved benchmark of single-cell CNV callers. SEQC2 HCC1395 / HCC1395BL, GRCh38. Expression arm (10x): inferCNV, CopyKAT, SCEVAN; allele arm (C1): Numbat.

All three expression callers reach AUROC ≥ 0.990 at every tumour fraction.

inferCNV is most stable, ≥ 0.997 across 1% to 80% (mean 0.998); band = CopyKAT replicate range.

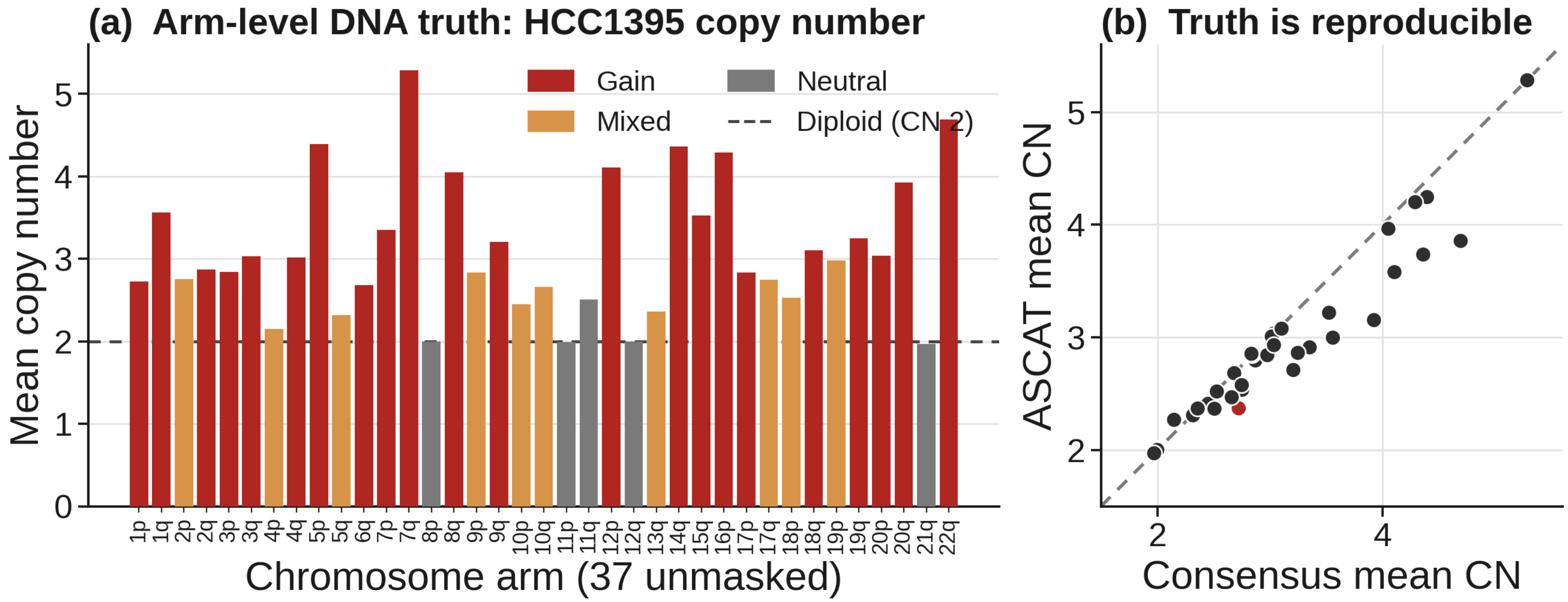


The benchmark rests on validated DNA truth

Arm-level truth from matched WGS: 22 gain, 10 mixed, 5 neutral of 37 unmasked arms.

Two orthogonal callers concur: consensus and ASCAT agree on 36 of 37 arms (97.3%).

Copy-number estimates track tightly between callers; only 1p differs (gain vs mixed).

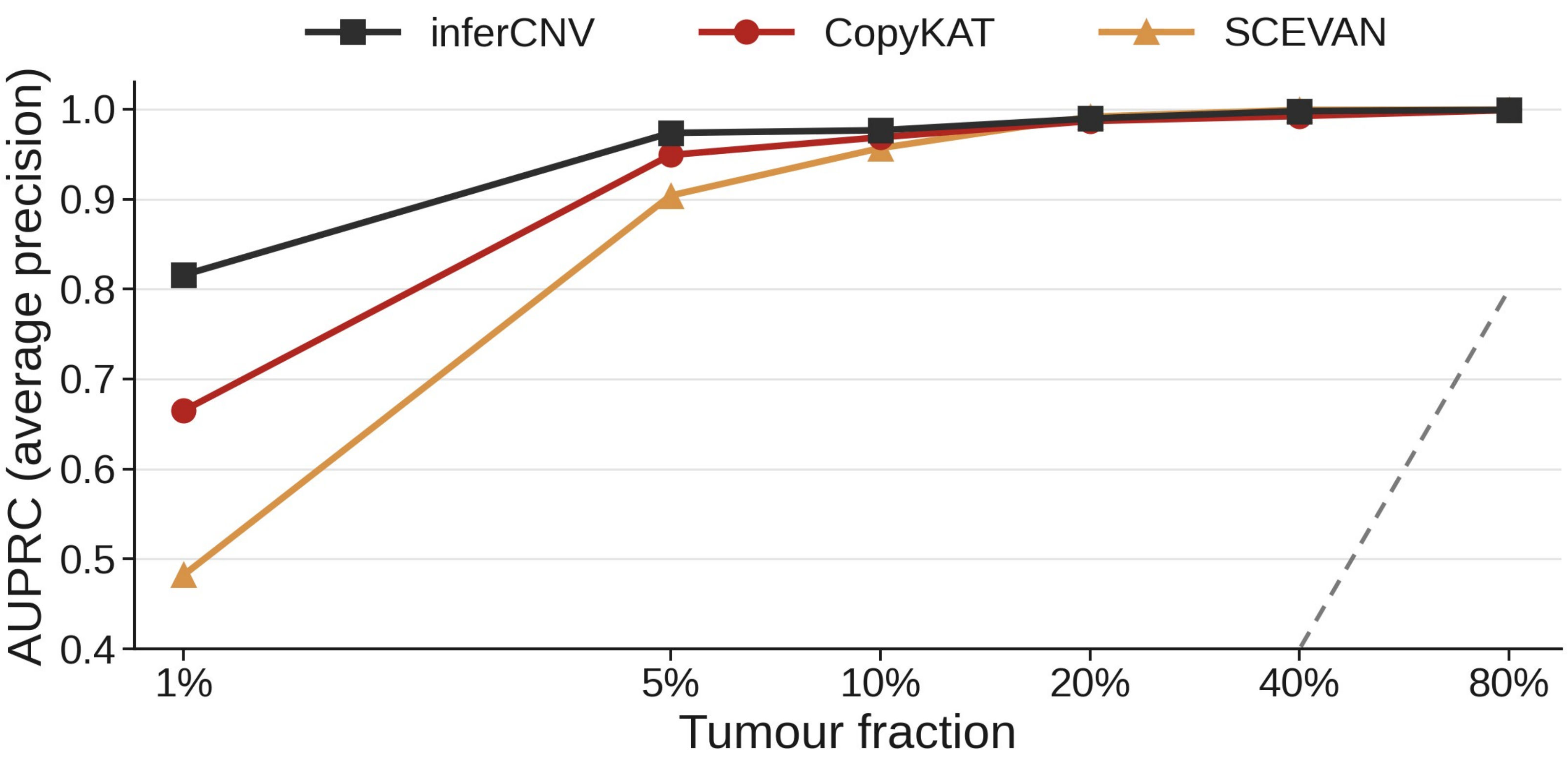


Detectability collapses at 1%, recovers by 5%

inferCNV holds precision-recall best at 1% purity (AUPRC 0.816).

SCEVAN falls hardest at 1% (0.482); CopyKAT intermediate (0.665).

All recover to ≥ 0.90 by 5% and ≥ 0.99 by 40%; dashed grey = AP chance.

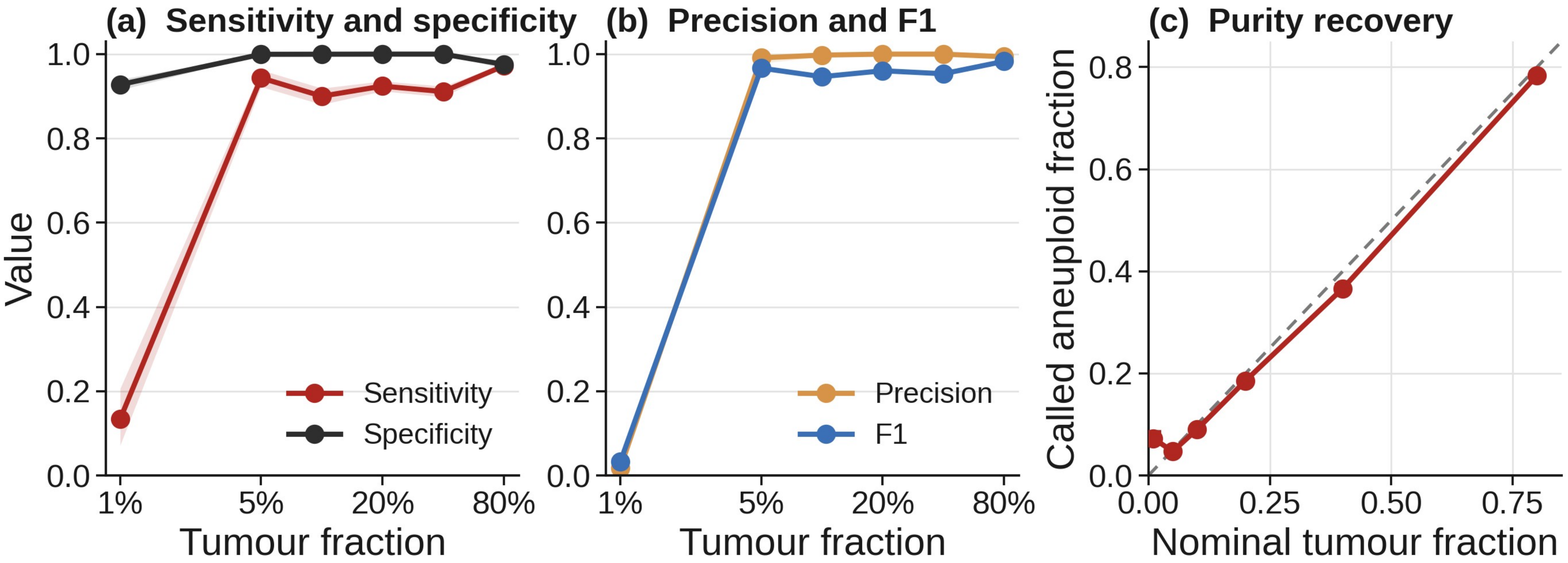


Decision metrics and purity recovery, with CIs

At the fixed threshold, sensitivity and precision are low at 1% but jump to ≥ 0.94 by 5% (F1 0.03 to 0.97).

Specificity stays ≥ 0.93 throughout; ribbons are cluster-bootstrap 95% CIs.

Called aneuploid fraction tracks nominal tumour fraction along the identity line.

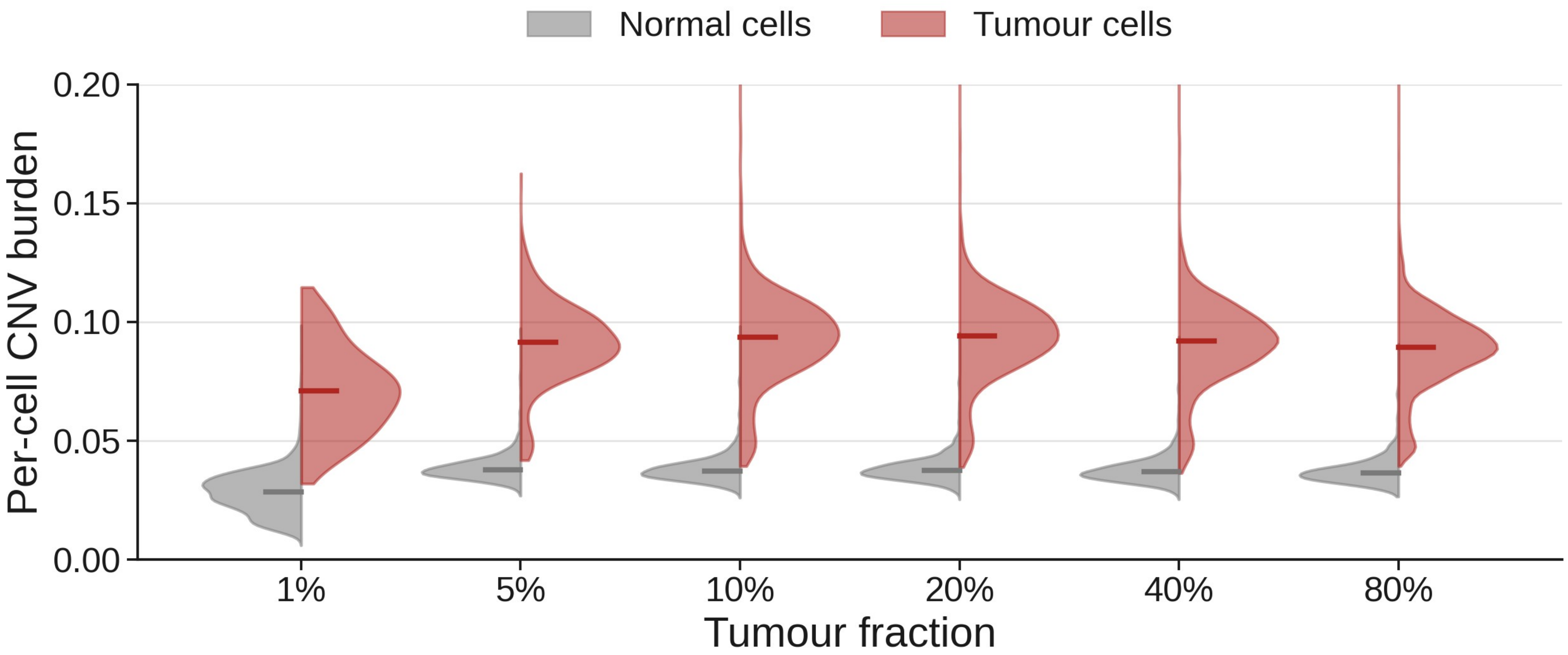


Malignant and normal cells separate by burden

Tumour-cell CNV burden sits well above normal at every purity, even at 1% (medians 0.071 vs 0.028).

This separation is why per-cell AUROC stays near 1.0 across the purity range.

Split violins pool 10 seeded replicates per fraction; median marked.

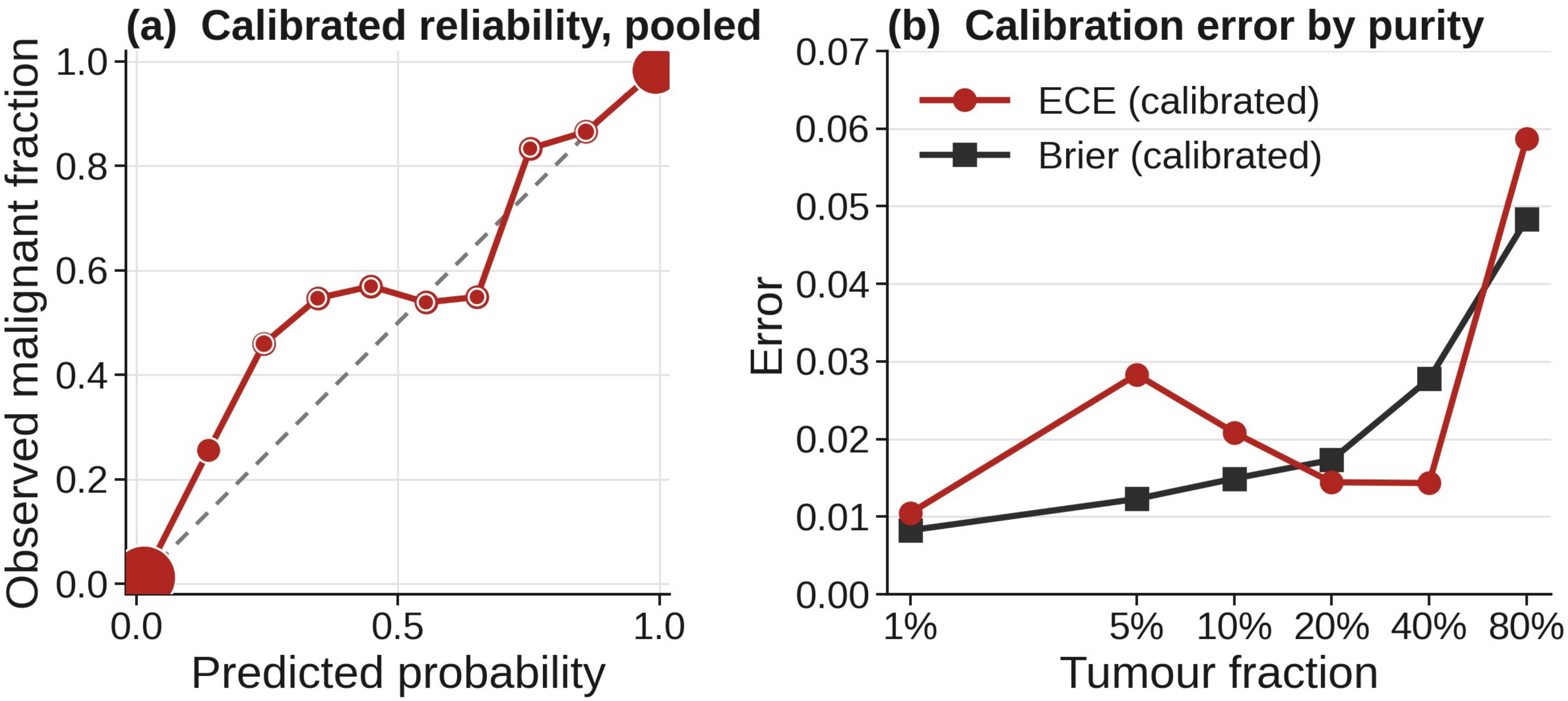


Probabilities are well calibrated

CopyKAT is the best-calibrated caller: pooled ECE 0.009, Brier 0.022.

Calibrated reliability tracks the diagonal across the probability range.

Per-fraction ECE stays ≤ 0.029 to 40%, rising to 0.059 at 80%.

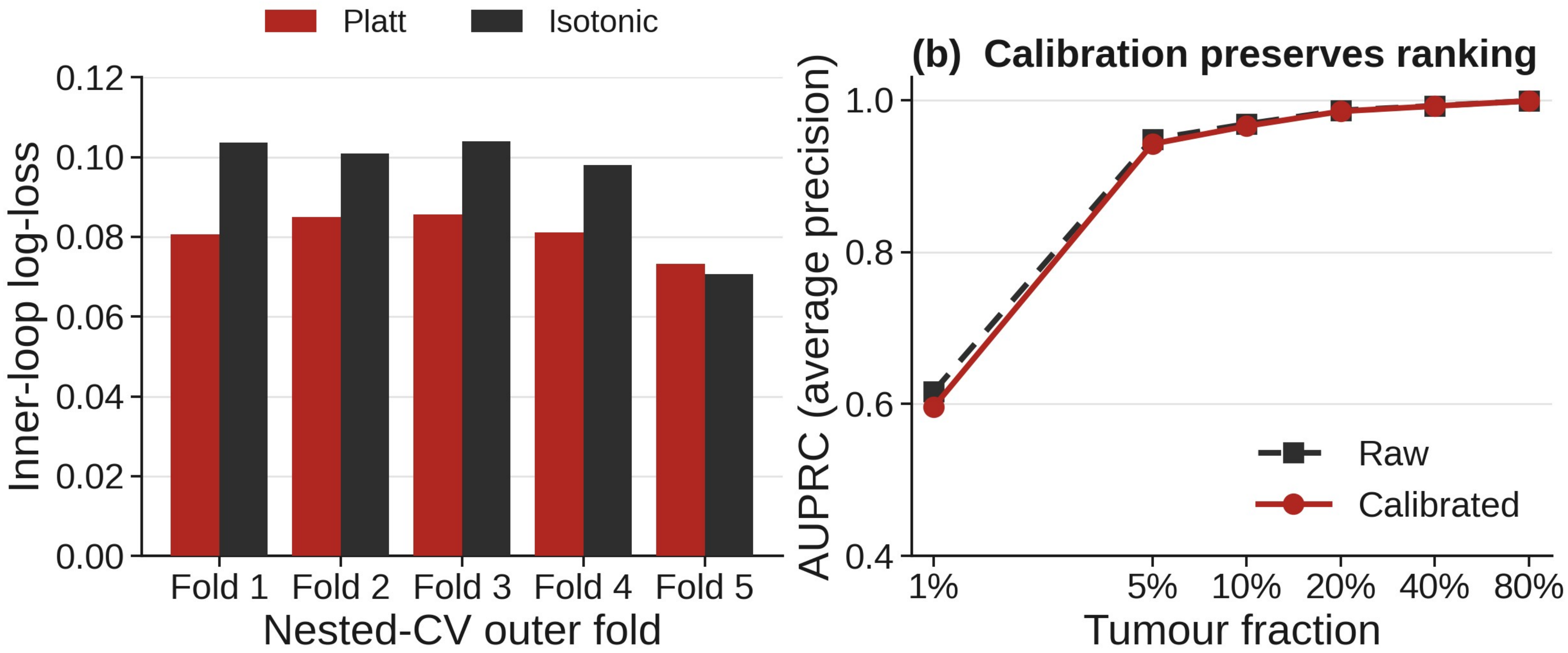


Calibration is chosen by group-aware nested CV

Platt scaling wins the inner loop in every outer fold; folds are grouped by source cell, so no cell leaks across folds.

Calibrated and raw AUPRC are nearly identical, so calibration preserves ranking.

Decision thresholds are set on training folds to target PPV 0.95 and NPV 0.98.

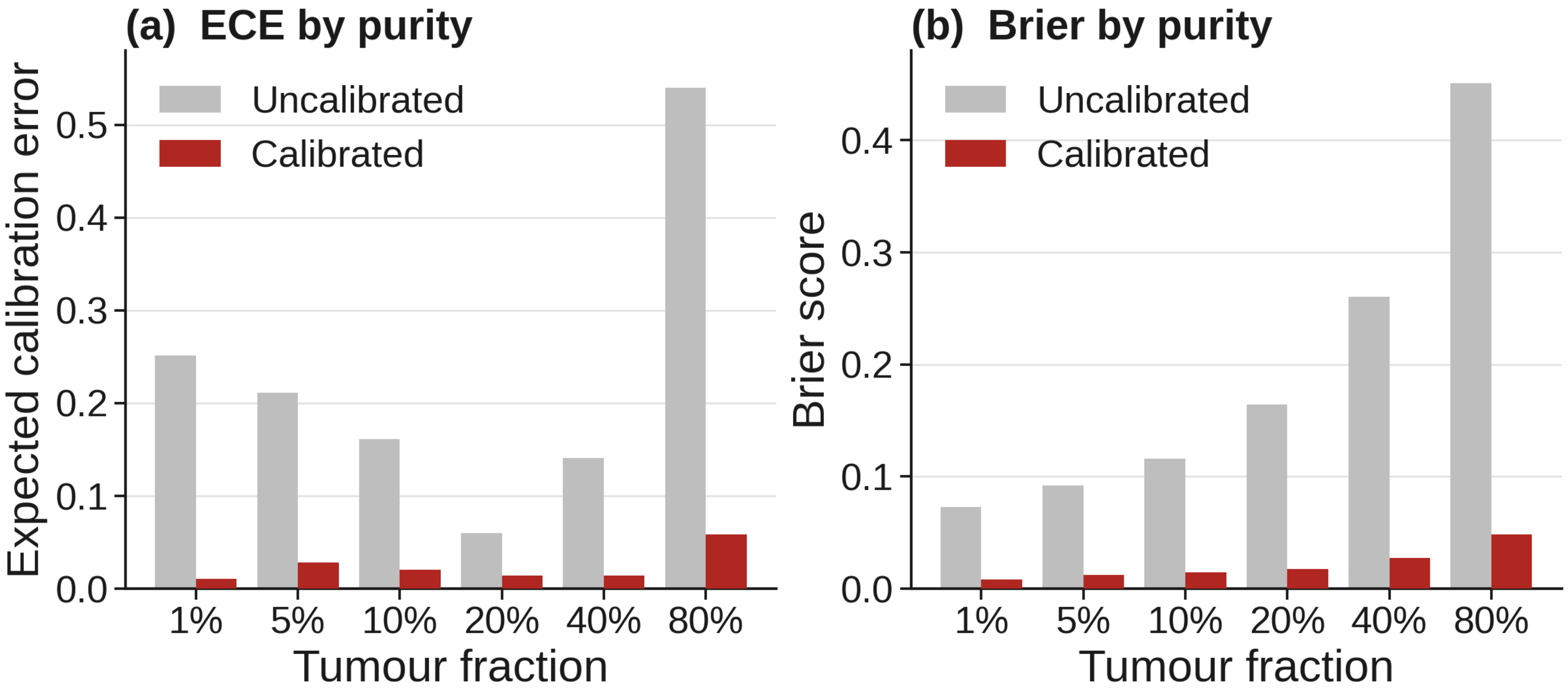


Calibration drift grows with purity

The uncalibrated score is overconfident, worst at 80% (ECE 0.54, Brier 0.45).

Calibration corrects it at every fraction; the gain is largest at 80% (Brier down 0.40).

After calibration, ECE and Brier stay low across 1% to 80%.

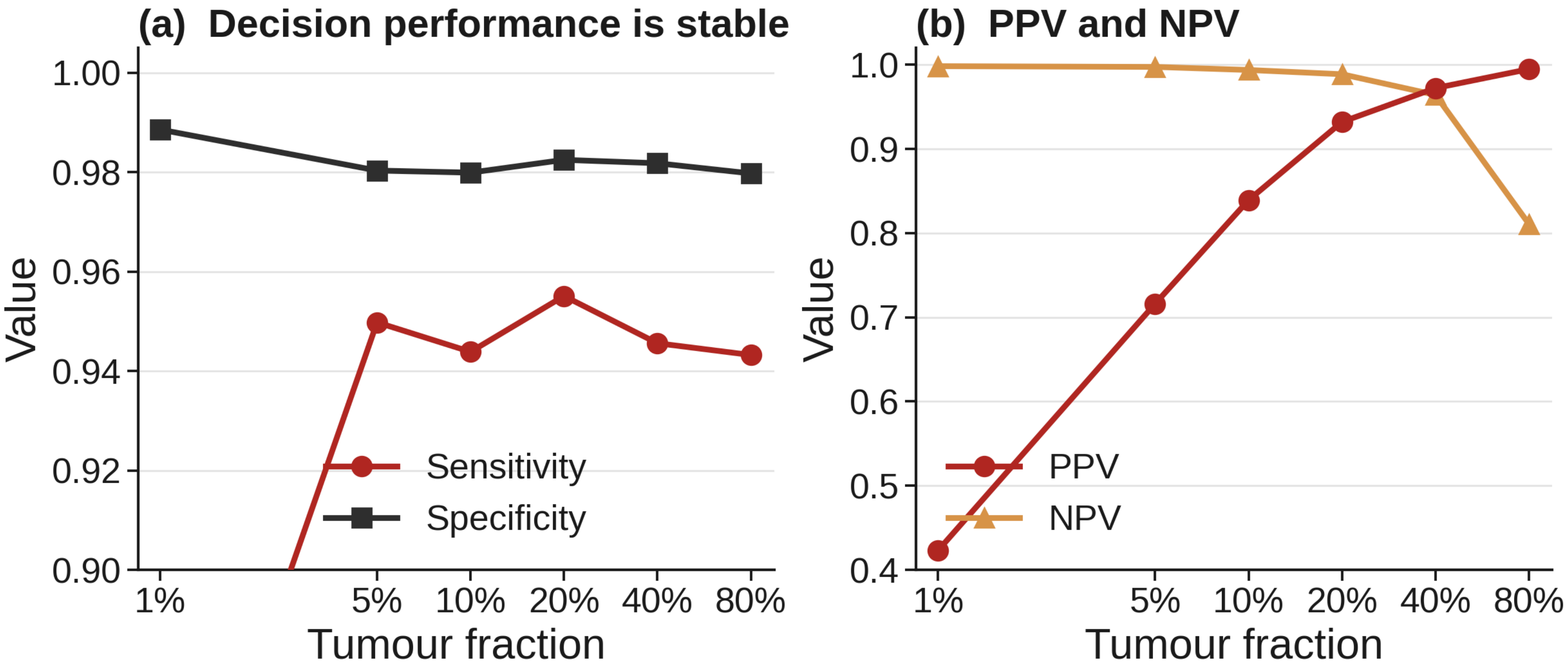


Calibrated decisions are stable across purity

Determinate sensitivity holds at ~0.95 and specificity at ~0.98 across 1% to 80%.

PPV rises 0.42 to 0.99 and NPV falls 0.998 to 0.81 with prevalence; a base-rate effect, not drift.

Every cell receives a determinate call; the indeterminate fraction is zero.

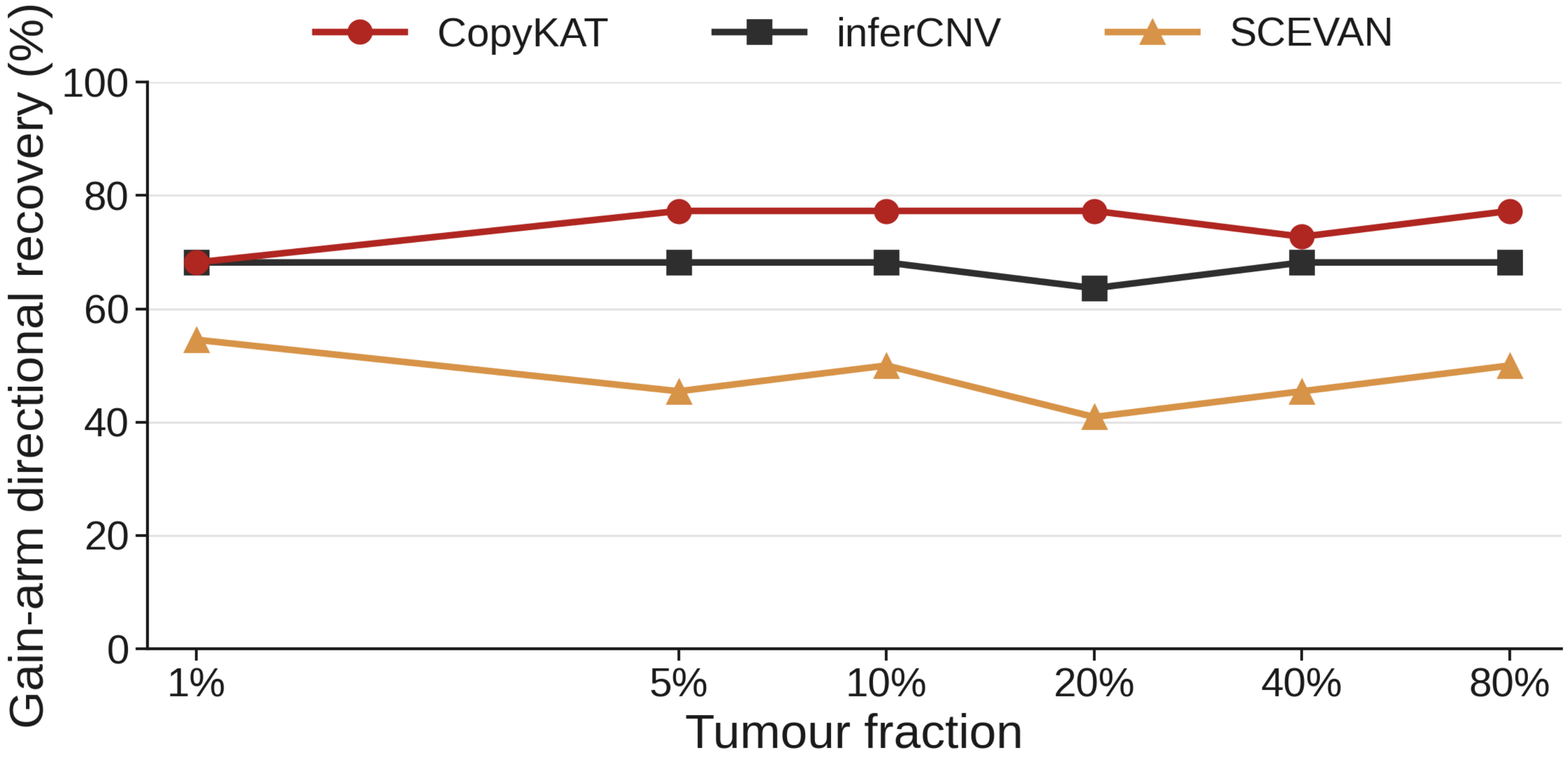


Gain-arm direction recovery by caller

CopyKAT recovers gain-arm direction best: 77.3% at 80% (17 of 22 gain arms).

inferCNV is flat at ~68.2% across all fractions; SCEVAN trails at 41% to 55%.

Loss is not evaluable: SEQC2 truth has no unmasked loss-dominant arms (22 gain, 0 loss).

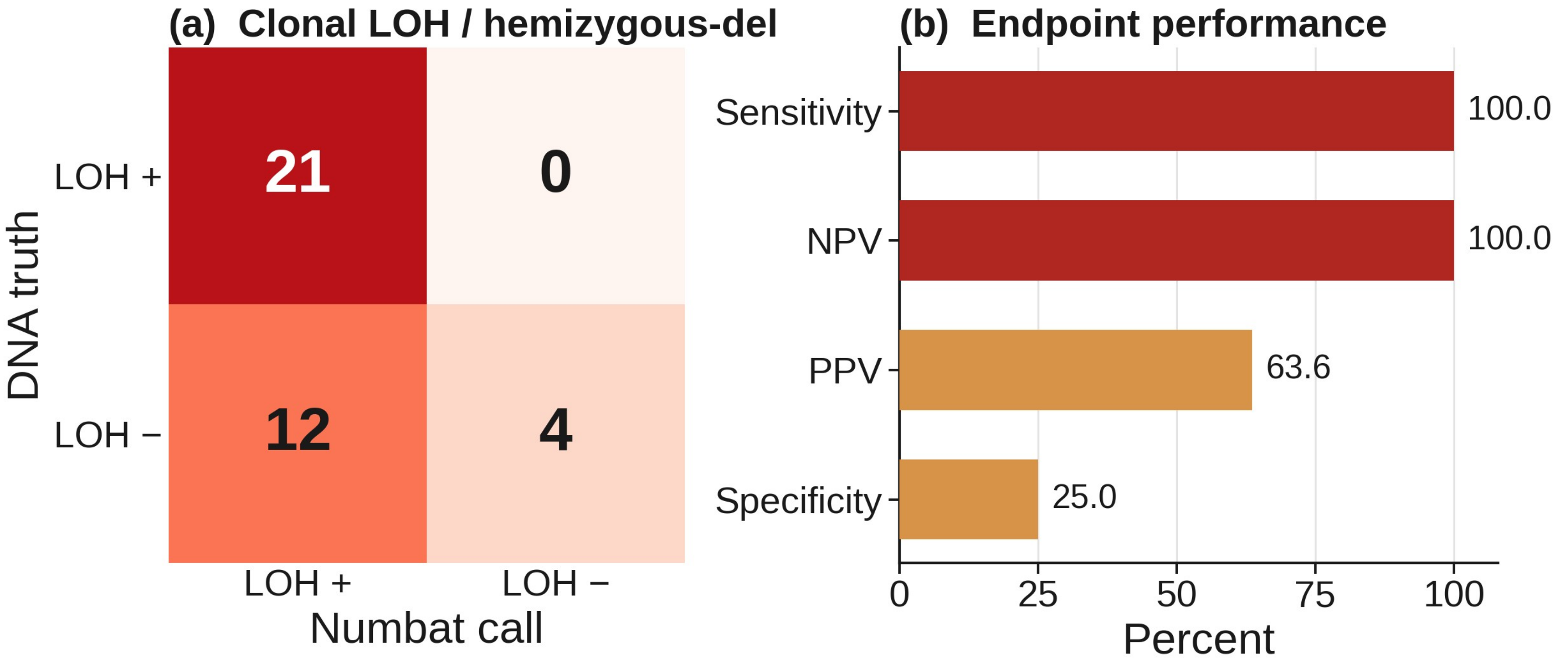


Haplotype-aware LOH concordance on C1

Per-cell malignancy is perfectly resolved on C1: AUROC 1.00 (expression, allele, joint).

Clonal LOH concordance is complete: 21 of 21 truth-LOH arms, including 17q (100% sensitivity).

The permissive LOH-or-del endpoint over-calls neutral arms (specificity 25%).

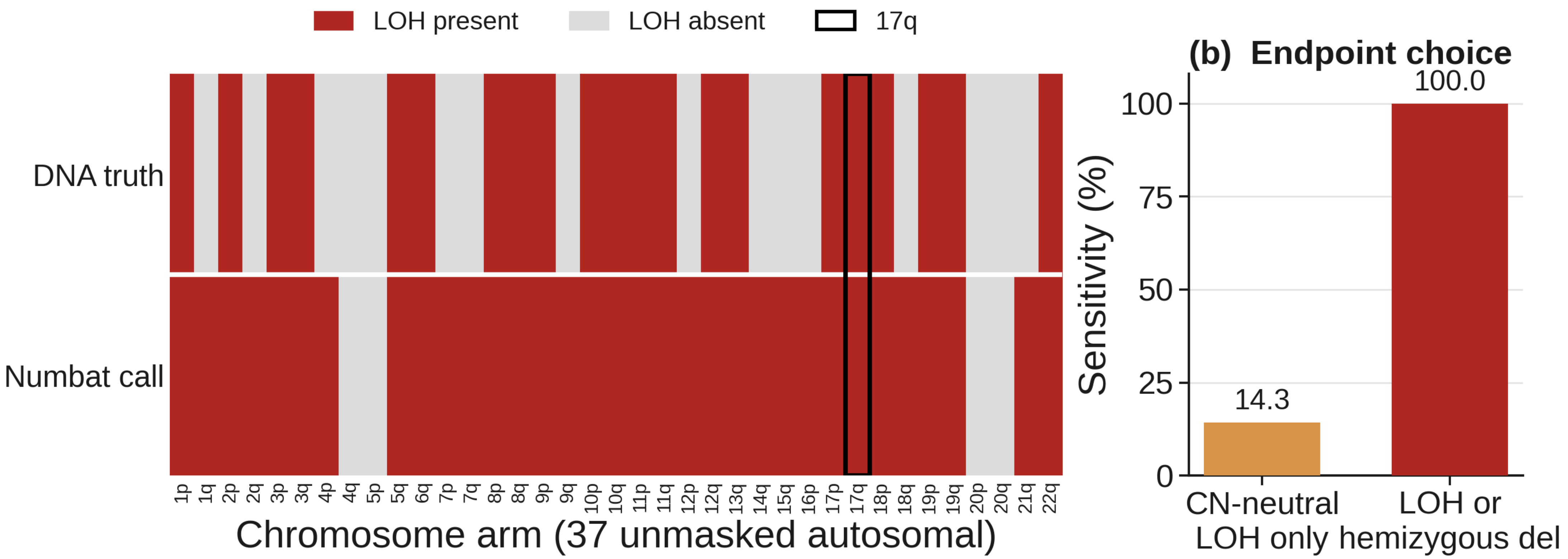


Numbat recovers LOH genome-wide, including 17q

All 21 truth-LOH arms are recovered, including 17q; the few over-calls fall on non-LOH arms.

The endpoint matters: LOH-or-hemizygous-deletion reaches 100% sensitivity, copy-neutral-LOH-only 14.3%.

HCC1395 loses heterozygosity largely through deletion, which the allele model captures.

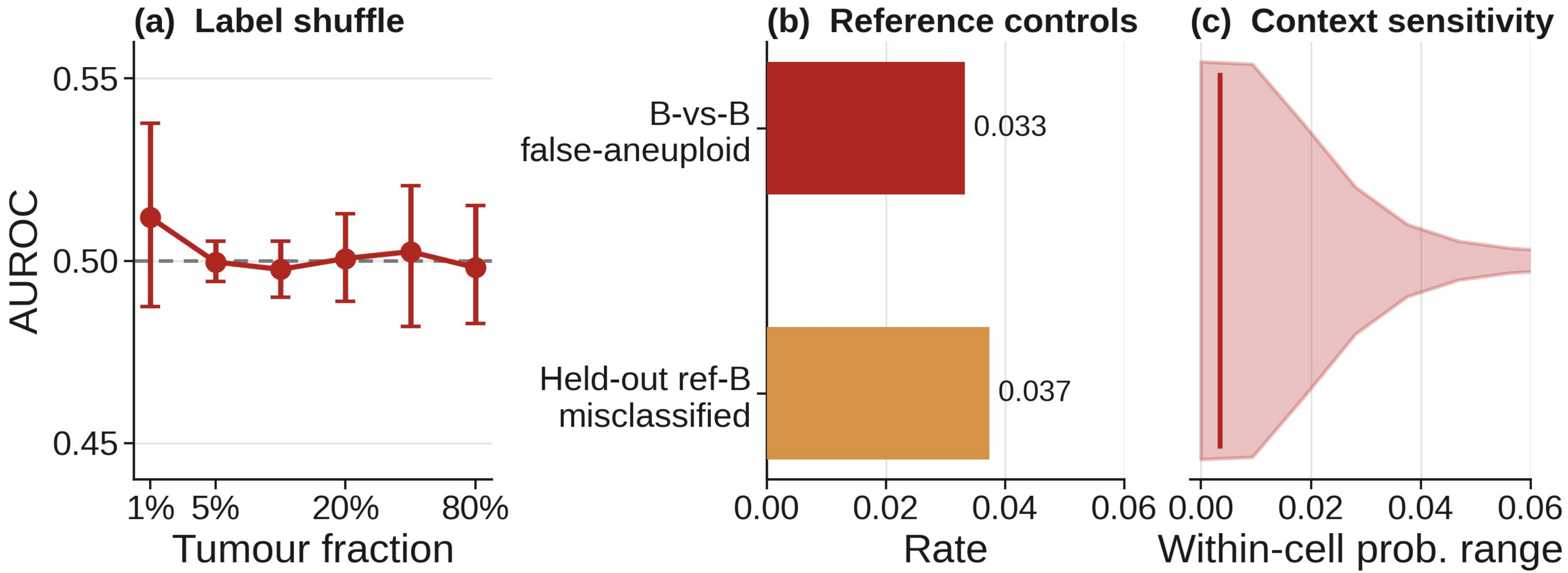


Negative controls all behave

Label-shuffle AUROC collapses to chance (~0.50) at every fraction.

B-vs-B false-aneuploid 0.033; held-out reference-B misclassification 0.037; barcode match 1.00.

Scores are stable to cohort: median within-cell probability range 0.003, 1.5% cross a boundary.

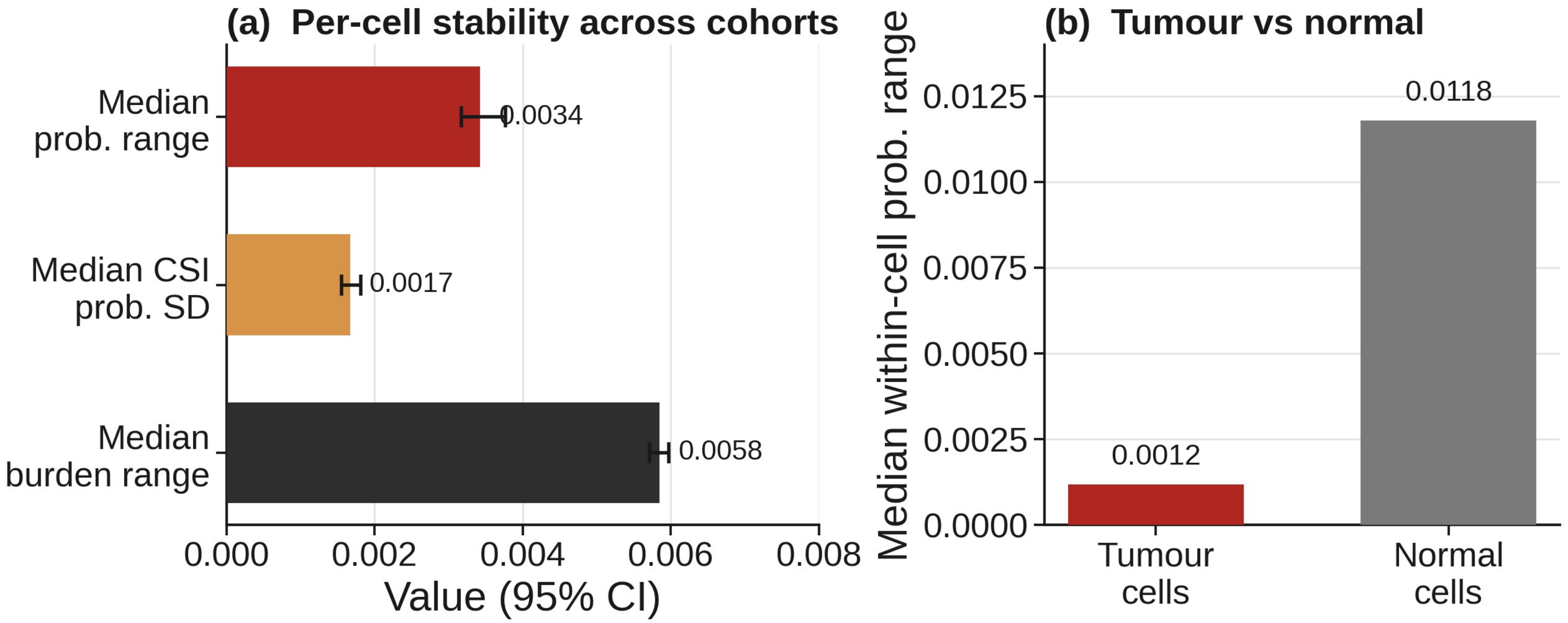


Per-cell scores are robust to cohort composition

Median within-cell probability range is 0.003 (95% CI 0.003 to 0.004); burden range 0.006.

Only 1.5% of cells cross a decision boundary across cohorts (95% CI 1.1% to 1.9%).

Tumour cells are the most stable (0.001); normal cells vary slightly more (0.012) but stay negligible.



Synthesis: best caller per axis

Axis	Metric	Best	Result
Discrimination	AUROC, per-cell	**inferCNV**	≥ 0.997 all fractions (mean 0.998)
Low-purity detectability	AUPRC at 1%	**inferCNV**	0.816 (≥ 0.90 by 5%)
Calibration	ECE / Brier, pooled	**CopyKAT**	0.009 / 0.022
Decision quality	Determinate sens / spec	**CopyKAT**	~0.95 / ~0.98 all fractions
Genomic fidelity	Gain-arm direction	**CopyKAT**	77.3% at 80% (17 of 22)
Allele / LOH (C1)	Clonal LOH concordance	**Numbat**	21 of 21 arms, 100% sensitivity
Per-cell (C1)	AUROC, expr / allele / joint	**Numbat**	1.00
DNA truth	Consensus vs ASCAT	**2 callers**	36 of 37 arms agree (97.3%)
Controls + context	Shuffle / B-vs-B / stability	**all pass**	0.50 / 0.033 / range 0.003

In progress: physical-mixture demultiplexing (experimental 5% and 10% mixtures) and the Nextflow pipeline.